Connecting to a Snapshot copy

Snapdrive for Unix

NetApp
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Connecting to a Snapshot copy

You can connect a Snapshot copy from one host to a different host.

SnapDrive for UNIX lets you connect a host to a Snapshot copy from a different location on a host. This new location can be on the host where you took the Snapshot copy (the originating host) or on a different host (the non-originating host).

Being able to set up the Snapshot copies in a new location means you can back up a Snapshot copy to another medium, perform maintenance on a disk group, or access the Snapshot copy data without disrupting the original copy of the data.

With this command, you can connect a host to a Snapshot copy that contains any of the following:

- LUNs
- A file system created directly on a LUN
- Disk groups, host volumes, and file systems created on LUNs
- NFS directory trees
- Disk groups, host volumes, and file systems on shared storage system

How the snapdrive snap connect command works

When you use the `snapdrive snap connect` command, it clones the storage for the entity you specify and imports it to the host:

- If you specify a Snapshot copy that contains a LUN (`-lun`), SnapDrive for UNIX maps a new copy of the LUN to the host. You cannot use the `snapdrive snap connect` command to specify a LUN on the same command line with other storage entities (`-vg`, `-dg`, `-fs`, `-lvol`, or `- hostvol`).
- If you specify a file system that resides directly on a LUN, SnapDrive for UNIX maps the LUN to the host and mounts the file system.
- If the source mount point is specified as the relative pathname in the `snap connect` command, SnapDrive for UNIX ignores the destination mount point specified in CLI and uses internal naming convention of the format `source_mount_point_<N>` to name the destination mount point.
- If you specify a Snapshot copy that contains a disk group, or a host volume or file system that is part of a disk group, the `snapdrive snap connect` command connects the entire target disk group. To make the connection, SnapDrive for UNIX re-activates all of the logical volumes for the target disk group and mounts all the file systems on the logical volumes.
- If you specify `autorename` option with the `snap connect` command, host volumes and file systems are always renamed. The disk groups are renamed only if they already exist on the host.
- If you specify a Snapshot copy that contains an NFS directory tree, SnapDrive for UNIX creates a clone of the FlexVol volume that contains the NFS directory tree. SnapDrive for UNIX then connects the volume to the host and mounts the NFS file system. Within the directory tree, SnapDrive for UNIX deletes any new NFS files or directories that you create after you created the Snapshot copy. SnapDrive for UNIX deletes any files or directories from the FlexVol volume that are outside the NFS directories that you connect, if the `snapconnect-nfs-removedirectories` configuration option is set to on.
- If you connect a Snapshot copy that contains NFS directory trees using the `-readonly` option, SnapDrive for UNIX mounts the Snapshot copy of the directory directly without creating a clone. You cannot use the

`snapdrive snap connect` command to specify NFS mountpoints on the same command line as non-NFS entities; that is, using the options `-vg`, `-dg`, `-fs`, `-lvol`, or `- hostvol`.

> The `snap connect` operations with `-split` option in vFiler environment are supported with Data ONTAP 7.3 and later.

Connecting Snapshot copies on mirrored storage systems

In the case of a Snapshot copy on a mirrored storage system, you can connect the Snapshot copy on the source storage system and the destination system.

When you create a Snapshot copy on a mirrored storage system, the Snapshot copy is automatically replicated, from the source system where it is created, to the destination (mirrored) storage system. SnapDrive for UNIX allows you to connect the Snapshot copy on the source storage system. You can also connect the Snapshot copy on the destination storage system.

Connecting multiple storage entities

You can connect a Snapshot copy containing multiple storage entities.

To connect a Snapshot copy that contains storage entities that reside on multiple destination storage systems you must meet the following requirements:

- The storage entities you specify at the command prompt must reside on a single storage system, or on a HA pair.
- The name of the volume of the source storage system must match the name of the volume of the destination storage system.
- You must set the *`snapmirror-dest-multiple-filervolumes-enabled`* variable in the snapdrive.conf file to “on”.

You can use one command to connect storage entities that reside on a single storage system or on a HA pair.

Snapshot connect and Snapshot restore operations

Snapshot copy clones the information when you connect to the Snapshot copy.

Unlike the Snapshot restore operation, the Snapshot connect operation does not replace the existing contents of the LUNs that make up the host entity with the Snapshot copy contents. It clones the information.

After the connection is established, both Snapshot connect and Snapshot restore operations perform similar activities:

- The Snapshot connect operation activates logical volumes for the storage entity, mounts file systems, and optionally adds an entry to the host file system table.
- The Snapshot restore operation activates the logical volumes for the storage entity, mounts the file systems, and applies the host file system mount entries that were preserved in the Snapshot copy.

Guidelines for connecting Snapshot copies

Follow the guidelines when connecting to Snapshot copies.

- The `snapdrive snap connect` command works only with Snapshot copies created in SnapDrive 4.2 for UNIX.
- On an originating host, SnapDrive for UNIX supports connecting and restoring Snapshot copies that are created by previous versions of SnapDrive for UNIX.
- For read and write access to NFS directory trees, the `snapdrive snap connect` command uses the Data ONTAP FlexVol volume feature, and therefore requires Data ONTAP 7.3 or later. Configurations with Data ONTAP 7.1 can connect NFS files or directory trees, but are provided with read-only access.
- If you set the *`enable-split-clone`* configuration variable value to “on” or “sync” during the Snapshot connect operation and “off” during the Snapshot disconnect operation, SnapDrive for UNIX does not delete the original volume or LUN that is present in the Snapshot copy.
- You have to set the value of Data ONTAP 7.2.2 configuration option vfiler. `vol_clone_zapi_allow` to “on” to connect to a Snapshot copy of a volume or LUN in a vFiler unit.
- The Snapshot connect operation is not supported on the hosts having different host configurations.
- The `snapdrive snap connect` command used to connect to a root volume of a physical storage system or a vFiler unit fails because Data ONTAP does not allow cloning of a root volume.

Information required for using the snapdrive snap connect command

To connect to a Snapshot copy, determine the type of storage entity, connect a Snapshot copy with the NFS directory tree to Data ONTAP 7.3 configurations, and so on.

The following table gives the information you need to supply when you use the `snapdrive snap connect` command.

> `snap connect` operations require Storage Foundation Enterprise License for vximport on AIX with Veritas.

Requirement/Option	Argument
Decide the type of storage entity that you want to use to attach the Snapshot copy and supply that entity's name with the appropriate argument. This is the value for the *src_fspec* argument. • If you connect a Snapshot copy of a LUN, SnapDrive for UNIX connects the LUN you specify. You cannot use the `-lun` option on the same command line with the `-vg`, `-dg`, `-fs`, `-lvol`, or `-hostvol` options. You can specify the short name of the LUN in the lun_name or qtree_name/lun_name format. • If you connect a Snapshot copy of a file system that is created directly on a LUN, SnapDrive for UNIX connects the LUN that has the file system. • If you connect a Snapshot copy of a disk group that has a host volume or file specification, the argument translates into a set of disk groups on the storage system. SnapDrive for UNIX connects the entire disk group containing the entity, even if the entity is a host volume or file system. • If you connect a Snapshot copy of an NFS file system, the argument translates to the NFS directory tree. SnapDrive for UNIX creates a FlexClone of the volume, removes directory trees that are not specified in the Snapshot copy, and then connects and mounts the NFS directory tree. If you specify an NFS mount point, you cannot specify non-NFS entities (`-vg`, `-dg`, `-fs`, `-lvol`, or `-hostvol`) on the same command line. SnapDrive for UNIX does not support symbolic links at the mount point level.	
LUN (`-lun` *file_spec*)	*short name of the LUN.*
The *s_lun_name* specifies a LUN that exists in the `-snapname` *long_snap_name*. The short lun_name is required. You cannot include a storage system or storage system volume name. The *d_lun_name* specifies the name at which the LUN is connected. The short *lun_name* is required. You cannot include a storage system or storage system volume name. You must specify a *d_lun_name*	
Disk group (`-dg` *file_spec*) or volume group (`-vg` *file_spec*)	*name of the disk or volume group*
File system (`-fs` *file_spec*)	*name of the file system*
Host volume (`-hostvol` *file_spec*) or logical volume (`-lvol` *file_spec*)	*name of the host or logical volume*
Connect a Snapshot copy with an NFS directory tree to Data ONTAP 7.3 configurations. • If your configuration uses Data ONTAP 7.3 or a later version of Data ONTAP with traditional (not FlexVol) volumes, you must specify this option to connect the Snapshot copy with readonly access (required). • If your configuration uses Data ONTAP 7.3 and later and FlexVol volumes, SnapDrive for UNIX automatically provides read-write access. Specify this option only if you want to restrict access to read-only (optional).	
-readonly	*set read only permission*

Requirement/Option	Argument
Optional: Supply a name by which the target entity is accessible after the storage entity is connected. SnapDrive for UNIX uses this name to connect the destination entity. This is the *dest_file_spec* argument. If you omit this name, the `snap connect` command uses the value you supplied for *src_fspec*.	
Name of target entity	*dest_file_spec*
Optional: Specify the names for the destination storage entities. If you included this information as part of the *dest_fspec/src_fspec* pair, you do not need to enter it here. You can use the `-destxx` options to specify names for destination storage entities if this information is not part of the *dest_fspec/src_fspec* pair. For example, the `-fs` option names only a destination mount point so you can use the `-destdg` option to specify the destination disk group. If you do not specify the name needed to connect an entity in the destination disk group, the `snapdrive snap connect` command takes the name from the source disk group. If you do not specify the name needed to connect an entity in the destination disk group, the `snap connect` command takes the name from the source disk group. If it cannot use that name, the operation fails, unless you included `-autorename` at the command prompt.	
Destination disk group (`-destdg`) or destination volume group (`-destvg`)	*dgname*
Destination logical volume (`-destlv`) or destination host volume (`-desthv`)	*lvname*
Specify the name for the Snapshot copy. Use the long form of the name where you enter the storage system name, volume, and Snapshot copy name.	
Snapshot copy name (`-snapname`)	*long_snap_name*
`-nopersist`	~

Optional: Connect the Snapshot copy to a new location without creating an entry in the host file system table.
 	
• The `-nopersist` option allows you to connect a Snapshot copy to a new location without creating an entry in the host file system table. By default SnapDrive for UNIX creates persistent mounts. This means that:
  ◦ When you connect a Snapshot copy on a AIX host, SnapDrive for UNIX mounts the file system and then places an entry for the LUNs that comprise the file system in the host's file system table.
  ◦ You cannot use `-nopersist` to connect a Snapshot copy that contains an NFS directory tree.	

`-reserve | -noreserve`	~

Requirement/Option	Argument
Optional: Connect the Snapshot copy to a new location with or without creating a space reservation.	
Igroup name (`-igroup`)	*`ig_name`*
Optional: NetApp recommends that you use the default igroup for your host instead of supplying an igroup name.	
`-autoexpand`	~
To shorten the amount of information you must supply when connecting to a volume group, include the `-autoexpand` option at the command prompt. This option lets you name only a subset of the logical volumes or file systems in the volume group. It then expands the connection to the rest of the logical volumes or file systems in the disk group. In this manner, you do not need to specify each logical volume or file system. SnapDrive for UNIX uses this information to generate the name of the destination entity. This option applies to each disk group specified at the command prompt and all host LVM entities within the group. Without the `-autoexpand` option (default), you must specify all affected host volumes and file systems contained in that disk group to connect the entire disk group. If the value you enter is a disk group, you do not need to enter all the host volumes or file systems because SnapDrive for UNIX knows what the disk group is connecting to. NetApp recommends that, if you include this option, you should also include the `-autorename` option. If the `-autoexpand` option needs to connect the destination copy of an LVM entity, but the name is already in use, the command fails unless the `-autorename` option is at the command prompt.	
The command fails if you do not include -autoexpand and you do not specify all the LVM host volumes in all the disk groups that is referred at the command prompt (either by specifying the host volume itself or the file system).	
`-autorename`	~
When you use the `-autoexpand` option without the `-autorename` option, the `snap connect` command fails if the default name for the destination copy of an LVM entity is in use. If you include the `-autorename` option, SnapDrive for UNIX renames the entity when the default name is in use. This means that with the -autorename option at the command prompt, the Snapshot connect operation continues regardless of whether all the necessary names are available. This option applies to all host-side entities specified at the command prompt. If you include the `-autorename` option at the command prompt, it implies the -autoexpand option, even if you do not include that option.	
`-split`	~
Enables to split the cloned volumes or LUNs during Snapshot connect and Snapshot disconnect operations.	

Requirement/Option	Argument
`mntopts`	~

Optional: If you are creating a file system, you can specify the following options:

- Use `-mntopts` to specify options that you want to pass to the host mount command (for example, to specify host system logging behavior). The options you specify are stored in the host file system table file. Allowed options depend on the host file system type.
- The *`-mntopts`* argument is a file system `-type` option that is specified using the mount command `-o` flag. Do not include the `-o` flag in the *`-mntopts`* argument. For example, the sequence -mntopts tmplog passes the string `-o tmplog` to the `mount` command, and inserts the text tmplog on a new command line.

> If you pass any invalid *`-mntopts`* options for storage and snap operations, SnapDrive for UNIX does not validate those invalid mount options.

Connecting to a Snapshot copy that contains LUNs

You can connect to a Snapshot copy that contains LUNs by using the `snapdrive snap connect` command.

Steps

1. Enter the following command syntax to connect to a Snapshot copy that contains LUNs:

   snapdrive snap connect -lun s_lun_name d_lun_name [[-lun] s_lun_name d_lun_name… -snapname long_snap_name [-igroup ig_name [ig_name…]] [-split]

   > The *s_lun_name* and *d_lun_name* should be in the format lun_name or qtree_name/lun_name.

SnapDrive for UNIX clones the LUNs you specify and connects them to a new location.

The following example connects the LUN mylun1, in hornet/vol/vol1/tuesdaysnapshot to mylun1copy:

```
# ./snapdrive snap connect -lun mylun1 mylun1copy -snapname
hornet:/vol/vol1:tuesdaysnapshot
connecting hornet:/vol/vol1/mylun1:
LUN copy mylun1copy ... created
(original: hornet:/vol/vol1/mylun1) mapping new lun(s) ... done
discovering new lun(s) ... done
```

The following example connects two LUNs, mylun1 and mylun2, to mylun1copy and mylun2copy, respectively:

```
# ./snapdrive snap connect -lun mylun1 mylun1copy -lun mylun2
mylun2copy -snapname hornet:/vol/vol1:tuesdaysnapshot
connecting hornet:/vol/vol1/mylun1:
LUN copy mylun1copy ... created
(original: hornet:/vol/vol1/mylun1)
mapping new lun(s) ... done
connecting hornet:/vol/vol1/mylun2:
LUN copy mylun2copy ... created
(original: hornet:/vol/vol1/mylun2)
mapping new lun(s) ... done
discovering new lun(s) ... done
```

Connecting to a Snapshot copy of storage entities other than LUNs

You can use the `snapdrive snap connect` command for connecting to a Snapshot copy that contains storage entities other than LUNs. This command cannot be used when destination names you provide are currently in use or, there is a file system name that is used as a mountpoint.

When you connect from a non-originating host to a Snapshot copy containing the VxFS file system mounted with the default mount `qio` option, you should have the Veritas license for Veritas File Device Driver (VxFDD) installed.

Steps

1. Enter the following command:

   snapdrive snap connect -snapname fspec_set [*fspec_set*…] -snapname *long_snap_name* [-igroup *ig_name* [*ig_name*…]] [-autoexpand] [-autorename] [-nopersist] [{-reserve | -noreserve}] [-readonly] [-split]

   In the preceding usage, *`fspec_set`* has the following format:

   {-dg | -fs | -hostvol} *src_file_spec* [*dest_file_spec*] [{-destdg | -destvg} *dgname*] [{-destlv | -desthv} lvname]

   This command must always start with the name of the storage entity you want to connect (for example, `-dg`, `-hostvol, or` `-fs`). If you specify an NFS mountpoint, you cannot specify non-NFS entities (`-vg`, `-dg`, `-fs`, `-lvol` or `-hostvol`) on the same command line.

SnapDrive for UNIX clones the LUNs you specify and connects them to a new location.

The following command line connects a disk group and uses the default names as the destination names (that is, it creates them from the source names):

```
# snapdrive snap connect -vg vg1 -snapname
filer1:/vol/vol1:vg1snapshot
connecting vg1:
LUN copy vg1_lun1_0 ... created
(original: filer1:/vol/vol1/vg1_lun1)
mapping new lun(s) ... done
discovering new lun(s) ... done
Importing vg1
```

The following command line connects a disk group with a single host volume. It also specifies a name for the destination host volume and disk group:

```
# snapdrive snap connect -lvol vg1/vol1 vg1copy/vol1copy -snapname
filer1:/vol/vol1:vg1snapshot
connecting vg1:
LUN copy vg1_lun1_0 ... created
(original: filer1:/vol/vol1/vg1_lun1)
mapping new lun(s) ... done
discovering new lun(s) ... done
Importing vg1copy
```

The following command line connects a disk group with two LUNs and two file systems. It specifies a destination name for each of the file systems, the host volume for one of the file systems, and the disk groups for both file systems:

```
# snapdrive snap connect -fs mnt/fs1 /mnt/fs1copy -destvg vg1copy \
-fs /mnt/fs2 /mnt/fs2copy -destlv vg1copy/vol2copy -destvg vg1copy
\ -snapname filer1:/vol/vol1:vg1snapshot
connecting vg1:
LUN copy vg1_lun1_0 ... created
(original: filer1:/vol/vol1/vg1_lun1)
LUN copy vg1_lun2_0 ... created
(original: filer1:/vol/vol1/vg1_lun2)
mapping new lun(s) ... done
discovering new lun(s) ... done
Importing vg1copy
```

The following command line includes the -autoexpand option as it connects a disk group with two file systems. It uses the default names as the destination names (that is, it creates them from the source names):

```
# snapdrive snap connect -lvol mnt/fs1 -snapname
filer1:/vol/vol1:vg1snapshot \
-autoexpand
connecting vg1:
LUN copy vg1_lun1_0 ... created
(original: filer1:/vol/vol1/vg1_lun1)
LUN copy vg1_lun2_0 ... created
(original: filer1:/vol/vol1/vg1_lun2)
mapping new lun(s) ... done
discovering new lun(s) ... done
Importing vg1
```

The following command line includes the `-autorename` option as it connects a disk group with two file systems and two LUNs:

```
# snapdrive snap connect -fs mnt/fs1 -snapname
filer1:/vol/vol1:vg1snapshot \
-autorename
connecting vg1:
LUN copy vg1_lun1_0 ... created
(original: filer1:/vol/vol1/vg1_lun1)
LUN copy vg1_lun2_0 ... created
(original: filer1:/vol/vol1/vg1_lun2)
mapping new lun(s) ... done
discovering new lun(s) ... done
Importing vg1_0
```

The following example connects to a Snapshot copy with file system, disk group created on LVM stack:

```
# snapdrive snap connect -fs /mnt/jfs1 /mnt/jfs1_clone -snapname
snoopy:/vol/vol1:snapLvm -autorename
connecting lvm1:
LUN copy lunLvm1_0 ... created
(original: snoopy:/vol/vol1/lunLvm1)
mapping new lun(s) ... done
discovering new lun(s) ... done
Importing lvm1_0
Successfully connected to snapshot snoopy:/vol/vol1:snapLvm
disk group lvm1_0 containing host volumes
jfs1_SdHv_0 (filesystem: /mnt/jfs1_clone)
```

In the following example, file system 1 (fs1) resides on storage system1, and file system 2 (fs2) resides on storage system1 and also on storage system2, which is the partner storage system. File system 3 (fs3) resides on storage system1, partner storage system 2, and storage system 3, which is not part of the HA pair. An

additional file system, fs4, resides entirely on storage system 4.

The following command creates a Snapshot copy of fs1, fs2, fs3, and fs4:

```
snapdrive snap create -fs /mnt/fs1 /mnt/fs2 /mnt/fs3 /mnt/fs4
-snapname fs_all_snap
```

The next command connect fs1 and fs2 on the destination storage system. Both fs1 and fs2 reside on a HA pair, so you can restore them with one command:

```
snapdrive snap connect -fs /mnt/fs1 /mt/fs2 -snapname fs_all_snap
```

The following command restores fs4:

```
snapdrive snap connect -fs /mnt/fs4 -snapname fs_all_snap
```

SnapDrive for UNIX cannot connect fs3 on the destination storage system, because this file system resides on storage system1, storage system 2, and storage system 3.

Connecting to Snapshot copies of shared storage entities other than LUNs

You can use the `snapdrive snap connect` command to connect to a Snapshot copy of shared storage entities other than LUNs.

Steps

1. Enter the following command:

   `snapdrive snap connect` *`fspec_set`* `[`*`fspec_set`*`…] -snapname` *`long_snap_name`*

   `fspec_set` **`[`*`fspec_set`*`…] -snapname`** ***`long_snap_name`*** **`[-devicetype shared] [-split]`**

   In this syntax, *`fspec_set`* `is: {-dg | -fs}_src_file_spec_ [`*`dest_file_spec`*` ] [-destdg `*`dgname`*`]`

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